

Mohammad Mehedi Hasan Akash, PhD
Postdoctoral Fellow in Mechanical Engineering
Joint College of Engineering
Florida A&M University - Florida State University, USA

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 🌐 [Website](#) | [LinkedIn](#) | [Google Scholar](#)

Teaching Interests

Ready to teach: Mechanical Vibrations; Fluid Dynamics; Computational Methods in Engineering; Computational Fluid Dynamics; Biofluid Mechanics; Engineering Mechanics

Willing to teach (with preparation): Design of Machine Elements; Engineering Materials; Manufacturing Processes; Heat Transfer; Measurements and Instrumentation Laboratory; Senior Design

New course development: Quantum Computing for Fluid Dynamics; Design for Additive Manufacturing with Simulation-Driven Design

Research Interests

Focus: Computational Fluid Dynamics (CFD) modeling for multiphase and turbulent flows; Quantum algorithms for fluid mechanics and transport

Specific Interests: Fluid Structure Interaction (FSI), BioFluid modeling, Multiphase modeling, Hemodynamics modeling using CFD

Academic Appointments

Postdoctoral Fellow , Mechanical Engineering, FAMU–FSU College of Engineering	07/2025 – present
Postdoctoral Fellow , Mechanical Engineering, Florida State University	04/2025 – 07/2025
Instructor of Record , Mechanical Engineering, South Dakota State University	2022 – 2023
Graduate Research Assistant , Mechanical Engineering, South Dakota State University	2021 – 2025

Education

Ph.D. in Mechanical Engineering

South Dakota State University, Brookings, SD, USA	01/2021 - 04/2025
Supervisor: Dr. Saikat Basu	
Focus: <u>Multiphase flow in complex biological systems</u>	

PhD Dissertation Title: Modeling Fluid Flow in Complex Biological Systems: Implications for Dense Tumor Microenvironments, Metastasis, and Bone Cancer | [Link](#)

M.S. in Mechanical Engineering

South Dakota State University, Brookings, SD, USA	01/2021 - 05/2023
Supervisor: Dr. Saikat Basu	
Focus: <u>Drug delivery and viral infection dynamics</u>	

M.S. Thesis Title: Modeling of Transport in Anatomic Respiratory Airways: Applications in Targeted Drug Delivery and Airborne Pathogenic Transmissions | [Link](#)

M.Sc. in Mechanical Engineering

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh	08/2018 - 12/2020
Specialization: Application of Fluid Dynamics	

B.Sc. in Mechanical Engineering

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh
Specialization: Application of Fluid Dynamics

02/2013 - 07/2018

Teaching Experience

Instructor of Record (sole instructor), Department of Mechanical Engineering, South Dakota State University

- **ME 323 Vibrations**, Spring 2023 — one section, 25 students
- **ME 323 Vibrations**, Fall 2022 — two sections, 27 students (15 + 12)
- Duties: designed and delivered all lectures, developed homework and examinations, graded student work, supervised a teaching assistant, maintained course site (D2L) with notes and solutions
- Student evaluations (IDEA): “excellent teacher” 4.6 and 4.7/5 (Fall 2022 sections), 4.1/5 (Spring 2023); 86% of 42 respondents rated the instructor an excellent teacher

Graduate Teaching Assistant, ME 446/546 Engineering Mechanics in Biomedical Applications, Fall 2023

- Duties: grading assignments

Mentorship

• Middle / High School outreach and other related activities (at SDSU)

1. SoDAK HUMANS CAREER Education Outreach (2024)
2. Dakota Dreams Summer Camp for middle school students (2022, 2023)
3. Youth Engineering and Technology Summer Camp for high school students (2023)

- Graduate students (SDSU, Basu Lab): M Yeasin | Topics: Inhaled smallpox transmission and transport in tumor microenvironments; co-author on *Scientific Reports* (2026) and *Frontiers in Pharmacology* (2026) articles. MT Hossain | Topic: Parametric modeling of intranasal spray devices; *Frontiers in Drug Delivery* (2025) article
- U.G. student (2021): M. Diab | Topic: Modeling particle filtration in face masks
- High school student (Summer 2022): A Tummala | Topic: Airborne smallpox transmission modeling; co-author on an APS DFD 2022 presentation and a *Scientific Reports* (2026) article

Research Experience

Postdoctoral Fellow

07/2025 – present

Computational and Theoretical Multiphysics Laboratory
Department of Mechanical Engineering, Joint College of Engineering
Florida A&M University - Florida State University, USA
Supervisor: Dr. Kourosh Shoele

Project 1: Fluid–Structure Interaction (FSI) — Investigating drag-reduction mechanisms inspired by dolphin skin using high-fidelity simulations (code is written in FORTRAN). Leveraged High-Performance Computing (HPC) resources, coupling CCNS (fluid solver) with TAHOE (solid solver).

Project 2: Quantum Algorithms for Fluid Mechanics — Developing quantum and quantum–classical hybrid algorithms for transport equations and homogenized multiscale turbulence modeling, including fixed-width quantum time-marching for advection–diffusion and variational reconstruction of velocity fields from sparse sensors; implemented and tested with the Qiskit quantum simulator (Python)

Postdoctoral Fellow

04/2025-07/2025

Computational and Theoretical Multiphysics Laboratory
 Department of Mechanical Engineering, Florida State University
 Supervisor: Dr. Kourosh Shoele

Project: Fluid–Structure Interaction (FSI) — Investigating drag-reduction mechanisms inspired by dolphin skin using high-fidelity simulations (code is written in FORTRAN). Leveraged High-Performance Computing (HPC) resources, coupling CCNS (fluid solver) with TAHOE (solid solver).

Graduate Research Assistant

01/2021 -04/2025

Research Topic: Modeling multiphase blood flow inside tumor microenvironment and CFD in respiratory airways - targeted drug delivery

Supervisor: Dr. Saikat Basu

Department of Mechanical Engineering, South Dakota State University, USA

Software used in PhD and MS : ANSYS Fluent, ICEM, Workbench, Multiphase simulation using Euler from ANSYS Fluent with **User Defined Function (UDF)**

Funding and Grantsmanship

NSF ACCESS Allocation| MCH250066

Turbulent Flow Interaction with Flexible Branched Canopies

Period: 10/2025 - 09/2026 | Estimated Allocation Value: \$25,372 | Role: Co-PI (wrote the proposal)

Award Resources: SDSC Expanse Projects Storage: 40,000.0 GB

SDSC Expanse CPU: 5,312,000.0 Core-hours

Publications and Presentations

Peer-reviewed journal publications

Note Underline denotes students supervised by Dr. Akash | * = Akash is corresponding author

8. **MMH Akash***, M Yeasin, S Mahmoudirad, RA Niloy, J Mohammad, K Reindl, A Pandey, S Basu
Reduced-order modeling of solute transport within physiologically realistic solid tumor microenvironment
Frontiers in Pharmacology, 17: 1746751, **2026** | [DOI](#)
7. M Yeasin, **MMH Akash**, A Malakar, AA Borojeni, A Tummala, J Wu, WD Bennett, WM Bodnar, JS Kimbell, A Chakravarty, JR Port, S Basu
Decoding the mechanophysiology for inhaled onset of smallpox with model-based implications for mpox spread
Scientific Reports, 16: 2142, **2026** (published online Dec 2025) | [DOI](#)
6. MT Hossain, A Malakar, M Yeasin, W O'Connell, **MMH Akash**, AA Borojeni, D Samanta, G Williams, J Reineke, G Farias, S Jung, J Suman, S Basu
Mechanics-guided parametric modeling of intranasal spray devices and formulations for targeted drug delivery to the nasopharynx
Frontiers in Drug Delivery, 5: 1721960, **2025** | [DOI](#)
5. A-B Merife, A Poudel, A Polshikova, ZJ Geffert, JA Horton, **MMH Akash**, A Pandey, S Basu, D Fougner, P Soman
3D osteocyte networks under Pulsatile Unidirectional Fluid Flow Stimuli (PUFFS)
ACS Biomaterials Science & Engineering, 11(10): 6216–6233, **2025** | [DOI](#)
4. **MMH Akash**, N Chakraborty, J Mohammad, K Reindl, S Basu
Development of a multiphase perfusion model for biomimetic reduced-order dense tumors

Experimental and Computational Multiphase Flow, 5(3): 319–329, **2023** | [DOI](#)

3. **MMH Akash**, Y Lao, PA Balivada, P Ato, NK Ka, A Mituniewicz, Z Silfen, J Suman, A Chakravarty, D Joseph-McCarthy, S Basu
On a model-based approach to improve intranasal spray targeting for respiratory viral infections
Frontiers in Drug Delivery, 3: 1164671, **2023** | [DOI](#)
2. J Yuk, **MMH Akash**, A Chakraborty, S Basu, LP Chamorro, S Jung
Morphology of pig nasal structure and modulation of airflow and basic thermal conditioning
Integrative and Comparative Biology, 63(2): 304–314, **2023** | [DOI](#)
1. S Basu, **MMH Akash**, NS Hochberg, BA Senior, D Joseph-McCarthy, A Chakravarty
From SARS-CoV-2 infection to COVID-19 morbidity: an in silico projection of virion flow rates to the lower airway via nasopharyngeal fluid boluses
Rhinology Online, 5: 10–18, **2022** | [DOI](#)

Preprints

3. **MMH Akash**, T Dev, NQ Nguyen, Y Chen, H Wu, K Shoele
Efficient Quantum Simulation of Variable-Coefficient Transport with Continuous Source Injection
arXiv:2608.27712 [quant-ph], **2026** | [Link](#)
2. NQ Nguyen, **MMH Akash**, K Shoele, Y Chen, H Wu
Reconstructing Fluid Velocity Fields from Sparse Sensors Using a Variational Quantum Algorithm
arXiv:2609.09268 [quant-ph], **2026** | [Link](#)
1. ML Tesch, A Malakar, SAA Rizvi, **MMH Akash**, G Ferrer, S Basu
Enhancing intranasal therapeutic penetration for treating respiratory illness: A clinical human factor feasibility study
preprints.org, **2023** | [Link](#) | submitted to *Journal of Patient Experience*

Peer-reviewed Conference Articles

1. M Yeasin, **MMH Akash**, A Malakar, A Chakravarty, S Basu
On modeling the mechanics of smallpox transmission through inhaled pathogens in respiratory domains
International Congress of Theoretical and Applied Mechanics (ICTAM), August 2024, Daegu, Korea

Conference Oral Presentations

Note: Lead author is the presenter

13. **MMH Akash**, T Dev, Q Nguyen, Y Chen, H Wu, & K Shoele
Quantum-Classical Hybrid Time-Marching Algorithm for a Linear Viscous Transport Model.
Florida Fluid Symposium V, May 2026, Gainesville, FL, USA
12. **MMH Akash**, T Dev, Q Nguyen, Y Chen, H Wu, & K Shoele
Quantum-Classical Iterative Homogenization Framework for Linear Multiscale Physical Systems.
American Physical Society (APS) Joint March Meeting and April Meeting - Global Physics Summit 2026, Denver, CO, USA
11. Q Nguyen, **MMH Akash**, T Dev, K Shoele, H Wu, & Y Chen
Variational Spacetime Simulation of the 1D Burgers Equation.
American Physical Society (APS) Joint March Meeting and April Meeting - Global Physics Summit 2026, Denver, CO, USA

10. **MMH Akash**, Q Nguyen, Y Chen, H Wu, & K Shoele
Toward Quantum-Native Multiscale Frameworks for Turbulent Flow Simulation.
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2025,
Houston, TX, USA.
9. **MMH Akash**, M Yeasin, A Pandey and S Basu
Physics-Based Modeling of Transport Phenomena in Biological and Bioinspired Systems
Florida Fluid Symposium IV, May 2025, Tampa, FL, USA
8. **MMH Akash**, M Yeasin, P Ran, A Merife, A Pandey, P Soman and S Basu
Modeling transport in physiologically realistic tumor microenvironment
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2024,
Salt Lake City, UT, USA
7. M Yeasin, **MMH Akash** and S Basu
Smallpox revisited: A mechanics-based model for its inhaled transmission
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2024,
Salt Lake City, UT, USA
6. **MMH Akash**, S Basu
Computational modeling of multiphase transport in physiologically realistic solid tumor vasculature and intratumoral domains
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2023,
Washington DC, USA
5. AAT Borojeni, **MMH Akash**, A Malakar, S Basu
An integrative modeling platform for smallpox transmission via respiratory routes
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2023,
Washington DC, USA
4. **MMH Akash**, Z Silfen, D Joseph-McCarthy, A Chakravarty, S Basu
Can Machine Learning predict particle deposition at specific intranasal regions based on computational fluid dynamics inputs/outputs and nasal geometry measurements?
SDSU Data Science Symposium, February 2023, Brookings, SD, USA
3. **MMH Akash**, A Tummala, S Basu
When fluid mechanics meets virology: revisiting smallpox – a modeling framework for its airborne transmission
American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2022,
Indianapolis, IN, USA
2. Z Silfen, **MMH Akash**, M Cherepashensky, A Chakravarty, S Basu, D Joseph-McCarthy
In Silico Prototyping for Intranasally Administered Agents for COVID-19 and Other Respiratory Pathogens
APS April Meeting, April 2022, New York, NY, USA

1. **MMH Akash**, A Mituniewicz, Y Lao, P Balivada, P Ato, N Ka, Z Silfen, A Chakravarty, D Joseph-McCarthy, S Basu
A better way to spray? – a model-based optimization of nasal spray use protocols
 American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2021, Phoenix, AZ, USA

Poster Presentations

7. **MMH Akash**, T Dev, Q Nguyen, Y Chen, H Wu, & K Shoele
A Fully Coherent Quantum Simulation of Nonlinear Transport Equation
 Florida Quantum Conference, April 2026, Tallahassee, FL, USA
6. T Dev, **MMH Akash**, Q Nguyen, Y Chen, H Wu, & K Shoele
Quantum-Inspired Tensor Networks for Three-Dimensional Turbulence
 Florida Quantum Conference, April 2026, Tallahassee, FL, USA
5. Q Nguyen, **MMH Akash**, K Shoele, Y Chen, & H Wu
Quantum Variational Simulation of the One-Dimensional Burgers Equation.
 Florida Quantum Conference, April 2026, Tallahassee, FL, USA
4. **MMH Akash**, M Yeasin and S Basu
Integrative modeling of solute transport phenomena in solid tumor microenvironments bridging computational and theoretical fluid dynamics
 SDSU Graduate Research, Scholarship and Creative Activity Day (GRSCAD) symposium, April 2024, Brookings, SD, USA
3. **MMH Akash** and S Basu
Theoretical and computational fluid mechanics modeling for transport in dense tumors
 American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2023, Washington DC, USA
2. **MMH Akash** and S Basu
Computational and theoretical fluid mechanics modeling for transport in dense tumors
 NIH-NIGMS The Centers of Biomedical Research Excellence (COBRE) Symposium, July 2023, Fargo, ND, USA
1. **MMH Akash**, N Chakraborty, S Basu
A multiphase tracking of perfusion through in silico dense tumor domain
 American Physical Society (APS) Division of Fluid Dynamics (DFD) Annual Meeting, November 2021, Phoenix, AZ, USA

Computational Resources

- High-Performance Computing (HPC) for CFD and FSI (MPI); NSF ACCESS systems (SDSC Expanse)
 - Scientific Software – ANSYS Fluent, ICEM CFD, Quirk, Simulink, SolidWorks, CFD-POST, FieldView, Workbench
 - Scripting Languages – Python (Qiskit), Wolfram Mathematica, MATLAB, ANSYS-APDL
 - Programming Languages – FORTRAN, C
 - Documentation / Graphics – $\text{\LaTeX}$, Adobe Illustrator, Inkscape, ImageJ
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Awards, Fellowships, & Grants

2022	Travel Grant, American Physical Society (APS)
2022	Bangladesh Sweden Trust Fund
2012	Higher Secondary Educational General Scholarship, Bangladesh Educational Board
2009	Primary General Scholarship, Bangladesh Educational Board

Media Coverage

- On **The Brookings Register**, dated 1-May-2023; [News Link](#)
Basu Lab applies fluid mechanics modeling to cancer research at South Dakota State University
Caption: Mohammed Mehedi Hasan Akash, a Ph.D. candidate at South Dakota State University, working with Saikat Basu, an assistant professor in the Department of Mechanical Engineering, on a project.
 - On **The Brookings Register**, dated 11-February-2022; [News Link](#)
Inhaling virus-laden droplets may increase COVID-19 severity
Caption: Using a model of the upper airway, doctoral student Mohammad Mehedi Hasan Akash, left, and assistant professor Saikat Basu of South Dakota State University's Department of Mechanical Engineering discuss adjustments to a computational aerosol transport model of how the novel coronavirus may spread from the nasopharynx into the lungs via aspiration of virus-laden droplets or boluses. Their results are published in Rhinology Online
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Experimentation Experience

- Performed table-top experiments testing liquid media penetration in 3D-printed nasal cavity casts
 - Experimented with a setup that I independently designed and arranged, administering sprays into a 3D-printed nasal cast at various angles to the horizontal to reflect common nasal spray usage guidelines
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Service

- **Manuscript Reviewer:**
 1. *Frontiers in Immunology*, section: Cancer Immunity and Immunotherapy
 2. *Frontiers in Bioinformatics*, section: Integrative Bioinformatics
 3. *PLOS One*
 - Member (08/2022 – Present): Biomedical Engineering Society (BMES) Chapter at SDSU
 - President (02/2022 - 05/2023): Bangladesh Student Association (BDSA) at SDSU
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References

Dr. Saikat Basu
(Ph.D. supervisor)
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Mechanical Engineering
South Dakota State University
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Dr. Kourosh Shoele
(Postdoc Supervisor)
Associate Professor
Mechanical and Aerospace Engineering
Florida State University
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Dr. Yucheng Liu
(Department Head, Ph.D. years)
Department Head & Professor
Mechanical Engineering
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